

Image Processing

Homework number 2

Due date: March 20 2025

1. Data-hiding techniques are used to hide a set of data in an image (or video sequence). In many applications it is essential that the hidden data be robust to various forms of attack (e.g., compression, filtering). In this problem we hide a binary image in a host image and evaluate the robustness of the technique to a special form of attack (JPEG lossy compression). Write a Matlab program to break the image “Einstein” into 8 bit planes. Replace the data in the 0-bit plane (least significant) with the binary image “camera man”. Compare the visual appearance of the image Einstein before data hiding and after it. Can you see any differences? Compress and decompress the modified Einstein image using JPEG for 4 different quality factors. For each quality factor after decompression decipher the hidden image (the 0-bit plane) and display it. Is it similar to the original binary image “camera man”? Quantify the difference by counting the number of pixels that are different between these two images. How changing the quality factor affects the number of pixels that are different? Is this technique robust to this form of attack (compression)? Repeat the same by replacing the “camera man” into the 1-bit plane and 2-bit-plane.

Hint: To compress an image using JPEG you can use the following commands in Matlab:

```
cn=c/255;
```

```
imwrite(cn,'filename.jpg','jpg','quality',qf);
```

Where qf is the quality factor and the image (c) is normalized before JPEG compression (cn). To read a JPEG compressed image you can use the following command:

```
d=imread('filename.jpg','jpg');
```

2. Problem 5.1 of the textbook.
3. Problem 5.10 of the textbook.
4. Problem 8.9 of the textbook
5. Download the gray scale image “Lena” from the website of the course. Take the 2-D FFT of the image. Then, in the Fourier domain, remove the phase (set the phase to zero). Take the inverse 2-D FFT of the image. Does the resulting image

look like the original image? why? Perform a low pass filtering on the Elaine image by leaving all Fourier coefficients that fall within the cutoff frequency of the low pass filter untouched and setting coefficients that fall outside to zero. After the low pass filter is applied, take the inverse 2-D FFT and view the resulting image. Try three different values for the cutoff frequency. Use Laplacian in the frequency domain and high-boost filter the image.

Submission: please save all your images in .tif or .gif format. Compress all your images into one file named xxxxxxquiz2.zip where xxxxxx is your student number and upload it to Avenue to Learn.